



ME 300:
Engineering Thermodynamics

Ideal Gas Equation of State: P - v - T

State principles

✓ equation of state

State = all the properties $\rightarrow \Phi_1 = f(\Phi_2, \Phi_3, \dots, \Phi_N)$

$\rightarrow$ Simple Compressible Substance state principle:

$\Rightarrow$ For S.C.S., the state is completely defined by two independent, intensive properties

$$\Rightarrow \boxed{\Phi_1 = f(\underbrace{\Phi_2, \Phi_3}_{\text{independent, intensive}})}$$

$$p, v \rightarrow p = \frac{1}{v}$$

State principles for simple compressible substances

$$P-v-T \text{ E.O.S.} \longrightarrow P = f_1(T, v) \quad v \sim [\text{m}^3/\text{kg}]$$

$$\begin{aligned} \text{Caloric E.O.S.} \longrightarrow u &= f_2(T, v) \\ h &= f_3(T, P) \end{aligned}$$

$$\begin{aligned} \text{Gibbs Relations} \longrightarrow S &= f_4(T, P) \\ s &= f_5(T, v) \end{aligned}$$



Think back to chemistry – what is an ideal gas?

Ideal gas – P-v-T equation of state

Ideal gas: $\boxed{Pv = RT}$

$$P \sim [Pa] \quad T \sim [K]$$
$$v \sim [m^3/kg]$$

↙ Particular gas const.

$$R = \frac{R_u}{M}$$

$$R_u = 8314.472 \frac{J}{kmol \cdot K}$$

$$M \sim [\underline{kg/kmol}]$$

$$P = \rho RT$$

$$\begin{aligned} \rightarrow P\vartheta &= MRT \rightarrow \vartheta \sim [m^3] \\ \rightarrow P\vartheta &= NR_u T \rightarrow N \sim [mol] \end{aligned} \Rightarrow M = \frac{M}{N}$$

$$\boxed{P\bar{v} = R_u T} \rightarrow \bar{v} \sim [m^3/mol]$$

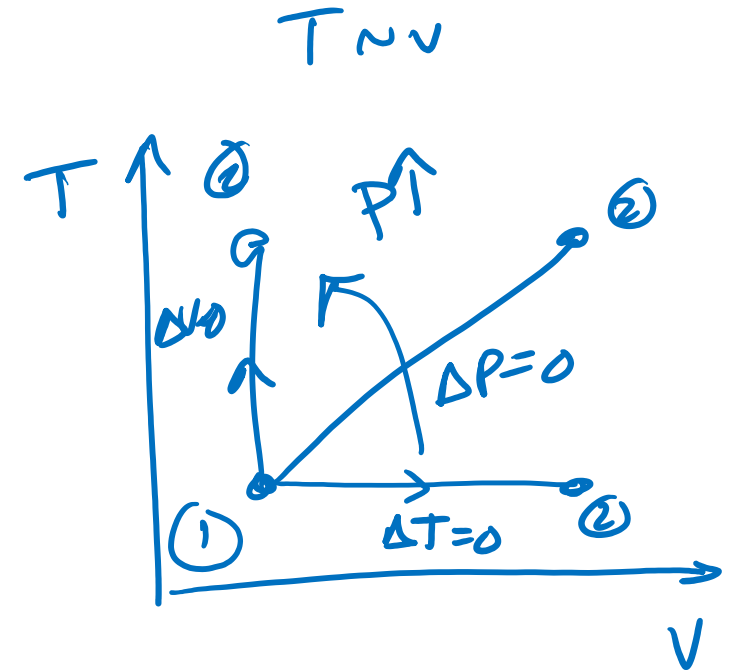
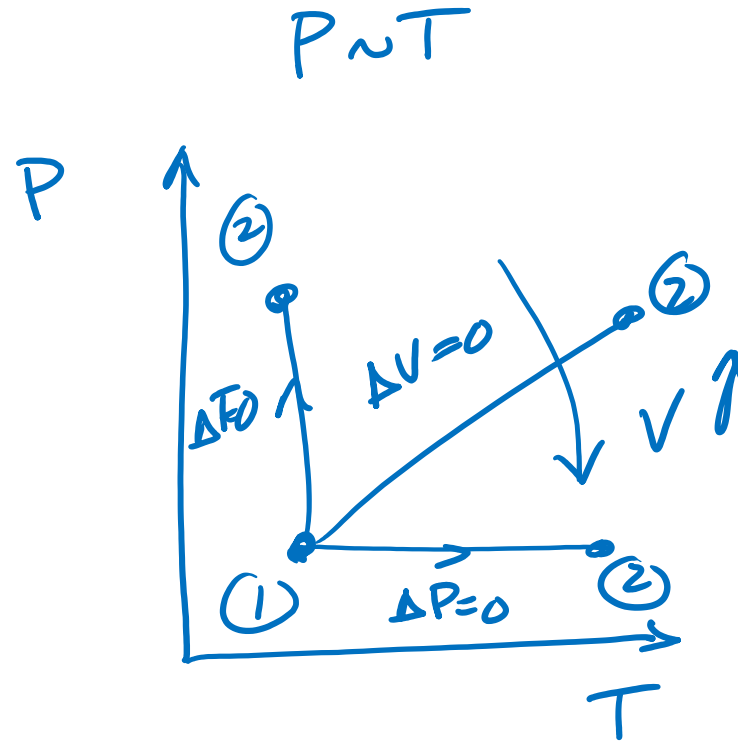
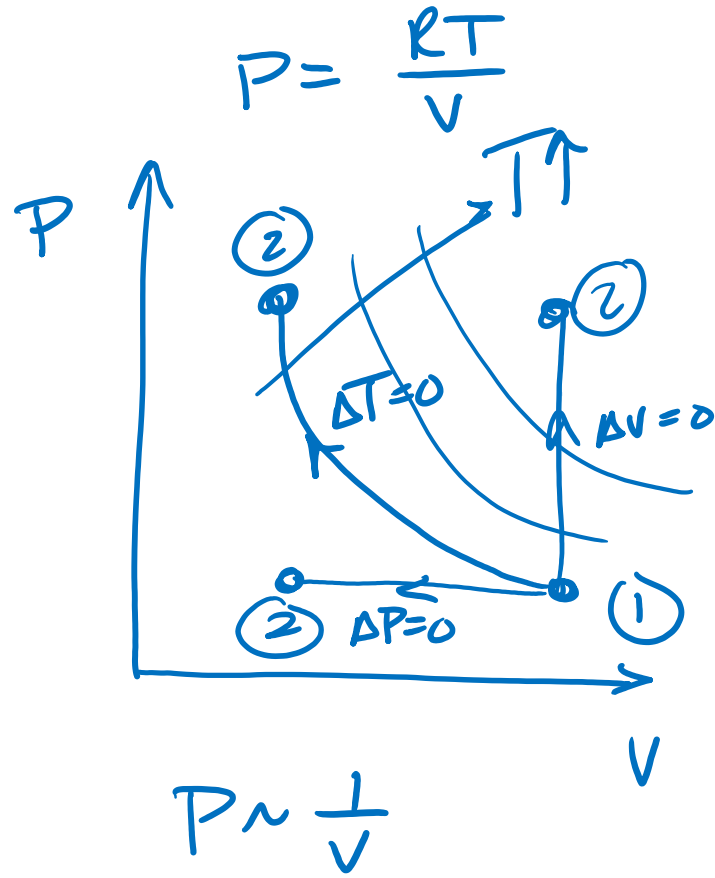
Ideal gas – Assumptions to derive P-v-T EOS

$$P\bar{v} = R_u T \rightarrow P\bar{v} = N R_u T$$

- ★ 1. molecules are far apart
 - $\bar{v}_{mol} \ll \bar{v}_{blw\ molecules}$.
 - collisions are infrequent
- 2. collisions are elastic
 - intermolecular forces are negligible

P-v-T equation of state – Plots

$$P_v = RT$$



P-v-T equation of state – Plots

